



Survival modeling of goal arrival times in English premier league

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Abstract

Prediction and modeling of association football (soccer) outcomes has gained increasing interest in the scientific community in recent years, both due to betting concerns and the need for a deeper understanding of the factors influencing soccer events. We introduce and examine the validity of a Bayesian model, which belongs to the class of accelerated failure time (survival) models and is characterized by its straightforward structure. We implement MCMC methodology to estimate the posterior summaries of the model parameters and suggest a novel algorithm that can be used to transform simulated goal arrival times into predicted goals. The proposed model achieves exceptional in-sample and out-of-sample performance by replicating the entire league in a remarkably precise manner and by making accurate predictions on the second half of the league using the first half as a training dataset. The structure of the proposed model is extendable, allowing for the inclusion of in-play covariates that can be used to further map the complex dynamics of soccer matches.

Keywords Goal arrival gap-time · Weibull distribution · Accelerated failure time models · Soccer

1 Introduction

Association Football (Soccer) has gathered increasing interest from researchers due to the great uncertainty involved in the final outcome of each game. Most of the football analytics articles focus of on the modeling/prediction of either the final outcome in the form of win/draw/loss using multinomial logistic regression models (Hvattum 2017) and other competing methods for the final score of the game using

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Poisson-based models and their extensions; see for example in Maher (1982), Dixon and Coles (1997), Karlis and Ntzoufras (2003, 2009). Here we are taking a different direction. We focus on modeling the goal arrival times within a football game.

Modeling the number of goals in a soccer game has been thoroughly studied over the last years. Maher (1982) was among the first to present a well-fitted and appealing Poisson regression model for such outcomes. Extensions of the basic double Poisson model have been proposed by Dixon and Coles (1997) and Karlis and Ntzoufras (2003) in order to account for the excess of zeros and for possible correlations between the goals scored by the two opponents. Additional models have been proposed to account for other characteristics of the model such as over-dispersion, a phenomenon which was first brought in the foreground in soccer by Karlis and Ntzoufras (2000) and then addressed by Karlis and Ntzoufras (2003). Another important issue in the use of dynamic time-dependent parameters used to capture the performance of the teams (Rue and Salvesen 2000; Owen 2011; Koopman and Lit 2015; Egidi and Gabry 2018). Additional research includes the article of Karlis and Ntzoufras (2009) in which a Bayesian version of the Skellam distribution is proposed as a model for the goal differences between the home and away team while Strumbelj and Šikonja (2010) introduces the notion of using the odds derived from the bookmakers as a forecasting tool by analyzing their effectiveness in 10,699 soccer matches. Additionally, the implementation of information drawn from historical data and bookmakers' odds has also been thoroughly researched by Egidi et al. (2018). Finally, for a comprehensive and detailed review of statistical and machine learning methods, see in Tsokos et al. (2019) and references therein. With regards to the modeling of goal arrival times that we deal with in this article, a very limited amount of research work has appeared in the relevant bibliography. One of the first related research papers is the one by Dixon and Robinson (1998) who have taken the goal arrival times to be coming from two Poisson birth processes when analyzing 4000 games from English competitions. They have concluded that the rate of the processes increases during the game and that it is influenced by the current score. Analysis of inter-arrival scoring times in Ice Hockey was conducted by Thomas (2007) using Weibull and Plateau-Hazard distributions. Finally, a direct approach to survival modeling in football was presented in Nevo and Ritov (2013) where the effect of the first and second goal was assessed using the Cox proportional hazards model in 760 Premier League games (2 seasons 2008–2010). This approach, although promising, is limited to the modeling of the first two goals independently of the scoring team, and by further considering the assumption of the proportionality of hazards. **Proportional Hazards Assumption**

Survival analysis models and the commonly applied Poisson models for predicting the final score capture different aspects of the data. Poisson models (and their extensions) are designed to model count data, meaning that in football, they answer the question: “*How many goals will a team score?*”. On the other hand, survival analysis models focus on modeling the time until an event (goal) occurs, rather than predicting the final outcome (Kleinbaum and Klein 2012). The final score is estimated as a byproduct of these models. Thus, survival models address the question: “*When will a goal be scored?*”. This can be particularly informative for in-play soccer (live) betting or for understanding the dynamics of

scoring frequency across opposing teams in a league. This discussion highlights that survival models require and model “richer” data as response inputs and, consequently, they answer a wider range of football-related questions compared to score-based models. Another argument for using survival models instead of Poisson-based models is that survival analysis allows for the estimation of hazard functions, which are crucial for understanding the actual risk of each event occurring (Cox 1972). Finally, most survival models (with the well-known exception of the exponential model) assume non-constant hazards, which implies that the rate of scoring varies over time. This is not the case for Poisson models, which are based on the less realistic assumption of constant hazards, as they imply the exponential distribution between goal arrival times (Andersen and Gill 1982).

In this work, we propose a general approach for modeling all goal inter-arrival times within a football match. We assess several assumptions such as the shape of the assumed distribution and potential dependence structures before we decide on the final proposed model. For this model, we assess the fit and the predictive performance of the selected model and it is compared with standard models such as the standard vanilla double Poisson model with respect to the prediction of the final score (although the primary aim of such model is to focus on the prediction of goal arrival times). Our survival analysis model models the inter-arrival times between subsequent goals. It is based on the following important assumptions: (a) when a team scores, then the scoring times of both teams restart to zero, (b) an event (goal) for a team defines the scoring time for the scoring team and a censored time of the opponent, (c) when no goal occurs (in the occasion of 0 – 0 or at the end of the game) then both times are considered as censored, and (d) all inter-arrival goal times are considered independent given the parameter values.

The implementation of our approach requires goal arrival times data that are not easily available via usual football data repositories such as <https://www.football.co.uk/>. For this reason, we were forced to explore other sources to extract the appropriate information about each of the matches considered. One of the richest electronic resources offering relevant information is the understat website (<https://understat.com/>), where the interested researcher can find an extensive amount of features of each game regarding major leagues such as the English Premier League, the Spanish La Liga and the German Bundesliga.

In what follows, this paper is organized into seven distinct sections. In Sect. 2, we formally explain the data structure and formulation. Section 3 provides modeling details and compares the fit for football data among the most popular parametric event-time models. Section 3.2 discusses various model extensions incorporating additional assumptions. It also compares these models using the Deviance Information Criterion (DIC) to assess their suitability for the given data. Section 4 focuses on the results providing parameter estimates, their practical interpretation, and the assessment of the goodness of fit of our model via simulations. Section 5 refers to additional modeling issues including the red card effect, model comparisons and censoring at half-time. We conclude this paper with a short discussion in Sect. 6 about the limitations of our approach and the possible future extensions.

2 Goal arrival times data in football

2.1 Data formulation

Modeling multiple event times in sports is not a trivial task since theoretically, an event can occur in any interval (no matter how small it is) with a positive probability. Hence, the actual time of the event is rarely captured precisely, and it is rounded up to the closest integer unit of time (that is, minute in football). Therefore, strictly speaking, the time t recorded for any goal scored is bi-variately censored, since it actually means that the goal occurred within the time interval $[t, t + 1)$. In our case, the times obtained from the box scores are indeed recorded with a precision of one minute.

In order to implement our approach we need to record the i goal arrival time by $\mathcal{T}_i$ which represents the periods (in minutes) between two subsequent events (goals). Moreover, the time between the last goal scored in a game and the end of the game is also recorded as a censored goal arrival time.

Therefore, the recorded intervals of each game with a score of $GO_1 - GO_2$ will be equal to the number of goals scored, plus one for the final goalless interval, that is, $GO_1 - GO_2 + 1$. The index $i \in \{1, \dots, n\}$ denotes the observation number, which corresponds to a recorded interval while n denotes the total number of recorded interval times under consideration and is set equal to the total number of goals scored plus the number of games considered in the dataset.

Additionally, to $\mathcal{T}_i$, we also use two binary indicators in order to specify the type of goal arrival interval:

- (a) the home team indicator $\mathcal{H}_i$ taking the value of one if $\mathcal{T}_i$ refers to the time of a goal scored by the home team or zero otherwise, and
- (b) a censoring indicator $\mathcal{C}_i$ which takes the value of one when the corresponding time refers to a censored time due to the end of the game and zero otherwise.

The indicator $\mathcal{C}_i$ is used to identify (for $\mathcal{C}_i = 1$) a time interval with no goal, which is censored for both opponent teams. This interval is observed once in each game and corresponds to the last recorded interval of each game. Moreover, when the game ends with no score (i.e., a 0–0 draw), it will be the only recorded interval.

Our approach is based on the idea that when a goal is scored, then the recorded time resets to zero, that is we model the gap arrival times. Hence, in our modeling approach, we assume two random variables T_{i1} and T_{i2} which reflect to the goal arrival times for the first and the second team in a game m . Then, we denote with t_{i1} and t_{i2} the corresponding i observed pair of times which are defined as

$$t_{i1} = \begin{cases} \mathcal{H}_i \mathcal{T}_i & \text{if } \mathcal{C}_i = 0 \\ \text{NA} & \text{if } \mathcal{C}_i = 1 \end{cases} \quad (1)$$

$$t_{i2} = \begin{cases} (1 - \mathcal{H}_i) \mathcal{T}_i & \text{if } \mathcal{C}_i = 0 \\ \text{NA} & \text{if } \mathcal{C}_i = 1 \end{cases} \quad (2)$$

The use of NA in the above data specification is due to the way we model censored observations in Bayesian analysis and the relevant MCMC software (i.e. in OpenBUGS or JAGS). Specifically, censored observations are considered as missing data where the censoring time point is known. Then, these (partially) “missing” data are generated from the corresponding (truncated) predictive distribution; see in Ntzoufras (Bayesian Modeling using WinBUGS page 297 - 298) for a detailed description.

To implement our approach in a Bayesian software such as OpenBUGS or JAGS, we needed to separately define the corresponding censoring times which were specified as

$$c_{i1} = (1 - \mathcal{H}_i)(1 - C_i)\mathcal{T}_i + C_i\mathcal{T}_i \quad (3)$$

$$c_{i2} = \mathcal{H}_i(1 - C_i)\mathcal{T}_i + C_i\mathcal{T}_i. \quad (4)$$

In the above formulation, if a team scores, we set its censoring indicator equal to zero (indicating no censoring) and the censoring indicator for the opposing team equal to the observed interval time (i.e., the time elapsed since the previous goal). However, if an end-of-game interval with no goals is observed, both censoring indicators will be set to the corresponding interval time.

Let us explain this by considering, as an example, the data layout for the first match in our dataset, between Manchester UND and Leicester, which ended with a score of 2-1. For this game, the recorded times are $\mathcal{T} = (2, 80, 9, 4)$, the home indicator values are given by $\mathcal{H} = (1, 1, 0, 0)$, and end-of-game censoring indicator takes values $\mathcal{C} = (0, 0, 0, 1)$. These data are re-arranged as shown in Table 1.

From Table 1, it is clear that the survival times are presented in an unusual manner. As we have already mentioned, an important assumption underlying our approach and therefore also the above structure the data is that the recorded event time resets after every goal. This data handling is equivalent to modeling the goal arrival times between each event. To be more precise, in the game of Manchester UND versus Leicester presented in Table 1 the the goals were scored in the 2nd, 82nd, 91st minute of the game with 5 min of added time due to delays (i.e. the final duration of the game was 95 min). If we denote by $\mathcal{T}^* = (0, 2, 82, 91, 95)$ the original

Table 1 Example of data layout for survival modeling: data refers to the first match of English premier league 2018–2019 (Manchester United – Leicester)

Game	Scoring time	Time interval	Home team	End-of-game censoring	Goal arrival times		Censoring times	
					Home	Away	Home	Away
	$\mathcal{T}_i^*$	$\mathcal{T}_i$	$\mathcal{H}_i$	$\mathcal{C}_i$	t_{i1}	t_{i2}	c_{i1}	c_{i2}
1	2	2	1	0	⇒ 2	NA	0	2
1	82	80	1	0	80	NA	0	80
1	91	9	0	0	NA	9	9	0
1	95	4	0	1	NA	NA	4	4

scoring times with by additionally appending the value of zero at the beginning of the vector and the total duration of the game, then the gap goal arrival times are simply calculated as $T_i = T_{i+1}^* - T_i^*$ for $i = 1, 2, 3, 4$. Hence, the value of $t_{21} = 80$ means that 80 min have passed since the previous goal was scored. The missing times (t_{41} and t_{42}) in the last line of the game (as it appears in Table 1) represents the inability to observe at what time would a team have scored from last scored goal due to the fact that the game was ended.

Finally, only one line of data will be recorded in the case of a 0–0 draw. The corresponding goal arrival times will be unobserved and thus will be set to NA, i.e., $(t_{i1}, t_{i2}) = (\text{NA}, \text{NA})$. Additionally, both censoring times will be set equal to the total duration of the game (90 min plus any extra injury or stoppage time added due to in-game delays). For instance, $(c_{i1}, c_{i2}) = (90, 90)$ for a game with no extra injury or stoppage time.

3 Bayesian weibull model formulation

3.1 A weibull vanilla model

Let us denote by T_{ij} the random variable that represents the goal arrival time concerning the i -th row and the j -th column (before observing it); where $i = 1, 2, \dots, n$ and $j = 1, 2$

$$\log(T_{ij}) = \mathbf{x}_{ij}^T \boldsymbol{\beta} + \sigma \epsilon_{ij} \quad (5)$$

where $\boldsymbol{\beta}$ is the vector of coefficients associated with the covariates $\mathbf{x}_{ij}^T$, σ is a scale parameter, and $\epsilon_{1j}, \dots, \epsilon_{nj}$ are independent and identically distributed according to the standard Gumbel distribution.

By setting $\gamma = 1/\sigma$ and making the relevant change of variables, it follows that

$$T_{ij} \sim \text{Weibull}(\gamma, \mathbf{x}_{ij}^T \boldsymbol{\beta}) \quad (6)$$

Hence,

$$T_{ij} \sim \text{Weibull}(\gamma, \lambda_{ij}), \quad \text{for } i = 1, 2, \dots, n, \text{ and } j = 1, 2,$$

with density $f(t) = \gamma \lambda t^{\gamma-1} e^{-\lambda t^\gamma}$, expected value $E(T) = \lambda^{-\frac{1}{\gamma}} \Gamma(1 + 1/\gamma)$, median $Md(T) = \lambda^{-\frac{1}{\gamma}} (\log 2)^{1/\gamma}$, and hazard function $h(t|\lambda, \gamma) = \lambda \gamma t^{\gamma-1}$ with $\gamma, \lambda > 0$.

The Weibull distribution is a more flexible counterpart to the Exponential distribution mainly due to the non-constant hazards. More specifically, the hazard is monotonically increasing (decreasing) for $\gamma > (<)1$ while for $\gamma = 1$ the Weibull distribution becomes identical to the Exponential distribution leading to constant hazard.

Moreover, we use the standard vanilla formulation adopted from the basic football models of Karlis and Ntzoufras (2003) given by

$$\begin{aligned}\log \{E(T_{i1})\} &= \mu + \text{home} + \text{att}_{HT_i} + \text{def}_{AT_i}, \\ \log \{E(T_{i2})\} &= \mu + \text{att}_{AT_i} + \text{def}_{HT_i},\end{aligned}$$

where HT_i and AT_i denote the home and away team for i observed event time.

This model can be also written in a log-linear form in terms of the λ_{ij} parameters given by

$$\begin{aligned}\log \lambda_{i1} &= \mu^* + \text{home}^* + \text{att}_{HT_i}^* + \text{def}_{AT_i}^*, \\ \log \lambda_{i2} &= \mu^* + \text{att}_{AT_i}^* + \text{def}_{HT_i}^*.\end{aligned}$$

In this model formulation, $\mu^* = -\gamma\{\mu - \Gamma(1 + 1/\gamma)\}$, $\text{home}^* = -\gamma \text{home}$, $\text{att}_k^* = -\gamma \text{att}_k$ and $\text{def}_k^* = -\gamma \text{def}_k$ for $k = 1, \dots, K$; where K is the number of teams in our dataset. Note, that the second formulation, which simply implements a vanilla structure on the λ parameters of the Weibull distribution rather on the expectations is just a re-parametrization of the first formulation.

Concerning the model parameters, μ is the intercept appearing in both Weibull distributions, the parameter home depicts the home effect, att_k and def_k are the parameters representing the attacking and defensive abilities of team k with $k = 1, 2, \dots, K$. For the attacking the the defensive parameters we need to consider a constraint in order to make the model identifiable. We have selected to use the sum-to-zero constraints given by

$$\sum_{k=1}^K \text{att}_k = \sum_{k=1}^K \text{def}_k = 0$$

to measure the differences in the log-expected times of scoring the next goal in comparison to a team of average attacking or defensive ability.

Parameter γ is essential for the behaviour of the goal-arrival times. When $\gamma < 1$ then the goal arrival rate decreases with time. Hence as the time passes the less likely it becomes to score. Accordingly, $\gamma > 1$ means that the goal arrival rate increases with time while for the value of one the distribution decays to the exponential distribution where the goal arrival rate is constant across time.

To complete the Bayesian formulation we need to specify our prior distributions for the model parameters. Here, we use low-information prior by considering normal priors with large prior variances for the parameters involved in the linear predictor, that is $\text{att}_k, \text{def}_k, \mu, \text{home} \sim \text{Normal}(0, 10^{-3})$ while for the positive parameter γ we use $\gamma \sim \text{Gamma}(10^{-3}, 10^{-3})$ distribution with mean equal to one and variable equal to 1000. In this work, we focus on using low-information prior distributions to allow the data to speak for themselves. The analysis could be further enhanced by incorporating expert knowledge or historical data into the prior, which is often available in football. However, we did not explore this aspect further, as the primary aim of this paper is to present and validate a model for goal arrival times by solely using our dataset. For a more detailed discussion on the use of prior information, see Sect. 6.

Details about the likelihood formulation for the model are presented in this section as well as for the estimation of the posterior predictive distribution using

MCMC samples can be found at Appendices A and B, respectively, which are available in the electronic supplement of this paper.

3.2 Model extensions and further assumptions

In this section, we indirectly check for the validity of two main assumptions of the Weibull model presented in Sect. 3.1: (a) the validity of the Weibull distributional assumption and (b) the independence between goal arrival times. For the first, in Sect. 3.2.1, we consider three alternative distributions which are popular in the survival analysis context. We fit them using the Bayesian approach and we compare them using the deviance information criterion (DIC) of Spiegelhalter et al. (2002). For the independence assumption, we use a bivariate distribution as well as random effects models and compare them with the independent Weibull vanilla model (Sect. 3.2.2).

3.2.1 Evaluation of different distributional assumptions for goal arrival times

In order to check for the appropriateness of the Weibull assumption we compare this model with other similar models using distributions which are also standard in survival analysis. Hence we compare the Weibull formulation introduced in Sect. 3.1 with three additional distributions/models: (a) the exponential distribution, (b) the log-normal distribution and (c) the log-logistic distribution; details about the distributions are given in Appendix C.1, available in the electronic supplement.

We compare and evaluate the predictive ability of the fitted models using DIC. Results are presented in Table 2 for the English Premier League 2018–19 data we consider in this article. The DIC values indicate that the Weibull model clearly outperforms its competing models in terms of the predictive performance.

3.2.2 Assessing dependence between goal arrival times

In this section, we introduce and assess the dependence between the goal arrival times of the two opposing teams by considering (a) the Marshall-Olkin Bivariate Weibull Model and (b) random effects models.

Among the popular choices for modeling bivariate survival times, is the Marshall Olkin Bivariate Weibull distribution. It is based on the Weibull distribution and can be assumed as a natural extension of it in order to account for dependence. The distribution can be derived by assuming independent Weibull random variables U_0, U_1

Table 2 Diagnostic criteria to assess each models' predictive performance and goodness of fit

	Distribution			
	Exponential	Log-normal	Log-logistic	Weibull
DIC	11,040	11,040	11,180	11,010

MCMC chain run for 10,000 iterations with 1000 iterations taken as burnin

and U_2 assuming that they have the same shape parameter γ and scale parameters λ_0 , λ_1 and λ_2 , respectively. Then for $T_1 = U_0 \wedge U_1$, $T_2 = U_0 \wedge U_2$, the assumed joint distribution of (T_1, T_2) is the Marshall Olkin Bivariate Weibull distribution. The formal details of this distribution are provided in Appendix C.2, available in the electronic supplement.

Moreover, we have introduced dependence using random effects. We have considered two random effects models: The first one accounts for effects concerning the game itself while the second was fitted by additionally considering team random effects. Details about the linear predictors of the fitted models can be found in Appendix C.3, available in the electronic supplement.

According to the DIC values in Table 3, the Weibull fixed-effects model outperforms the game-only random-effects models. To explore potential negative correlations between goal arrival times, a reversed-signs random-effects model (labelled as “Different signs” model in column 5 of Table 3) was examined. For the random-effects model incorporating team effects, the DIC is identical to that of the fixed-effects model, indicating no improvement in the predictive performance.

4 English premier league 2018–19 data

The English Premier League 2018–2019 data has received a reasonable amount of attention as it is considered a major annual soccer event. This data consisted of 20 teams and $n_G = 380$ games in total while the total number of observations following the data structure presented in Table 1, was $n = 1452$ (source: English Premier League understat webpage: <https://understat.com/league/EPL>). Among the 380 games, 48% (34%) were home (away) wins while 19% were draws. Event time-focused descriptives include the median scoring time of the home (away) team which was measured to be 50 (51) with a standard deviation of 27 (26) implying a rough empirical estimate of the median arrival time driven score for this data to be $1 - 1$.

4.1 Model based inference

The model was fit to the data using Markov Chain Monte Carlo (MCMC) via R2OpenBUGS software and it ran until convergence was assured (11,000 iterations with 1000 iterations as burn-in).

Table 3 Diagnostic criteria to assess each models predictive performance and goodness of fit

	Model				
	Independent	Bivariate	Random effects		
	Weibull	MOBW	Game effects	Different signs	Team effects
DIC	11,010	13,910	11,150	11,150	11,010

MCMC chain run for 11,000 iterations with 1000 iterations taken as burnin

Table 4 Posterior summaries for the 2018–19 English premier league under the Weibull fixed effects model - the log(Mean) case

	Mean	Median	SD	Posterior quantiles	
				2.5%	97.5%
Home advantage (<i>home</i>)	-0.193	-0.199	0.053	-0.286	-0.086
Intercept (μ)	4.025	4.028	0.035	3.952	4.090
Weibull shape (γ)	1.131	1.130	0.028	1.078	1.186

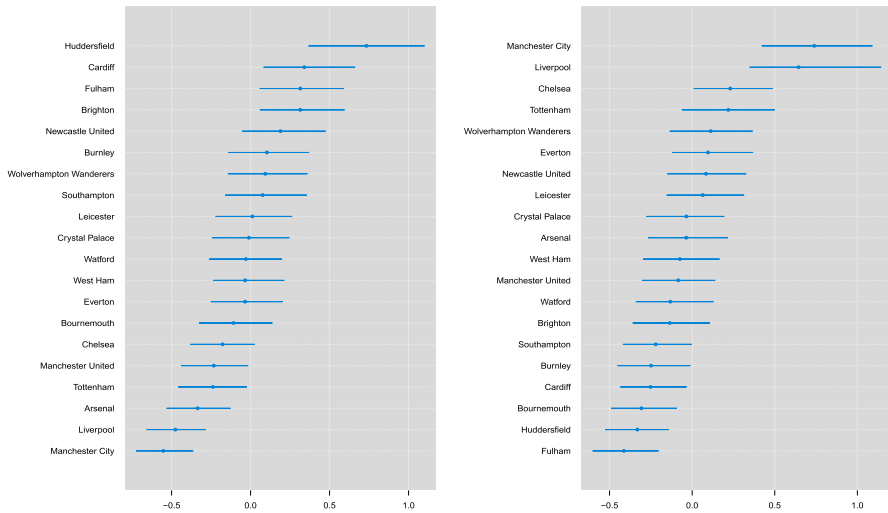


Fig. 1 95% Posterior intervals for the attacking (left) and defensive (right) abilities of each of the teams in the dataset along with posterior estimates of each of the teams' final points using both the posterior mean and the posterior median (25,000 iterations, 1000 burn-in)

Starting from estimates regarding the attacking and defensive abilities of each team, the posterior means accompanied with forests plots of the posterior team abilities are presented in Fig. 1 while the estimates regarding the remaining parameters of the model are given in Table 4.

4.2 Interpretation of the model

This section focuses on the analysis and the interpretation of the estimated model parameters for fixed effects Weibull model fitted on the data from the matches of the English Premier League, season 2018/19; see Sect. 3 for the detailed description of the proposed full Bayesian formulation.

Table 4, presents the basic model parameters of the fitted model. From this table, we observe that both the home effect *home* and the constant parameter μ appear to be away from zero indicating that should be included in the model formulation. Specifically, there is a clear home advantage (posterior mean of -0.19)

suggesting that the home team experiences a decrease in mean or median goal arrival time of -0.19 (on a logarithmic scale) and a 17.3% reduction in median goal arrival time (on the original scale) compared to the corresponding goal arrival times when playing away against the same team (posterior credible interval: $8.25 - 25\%$). Concerning the constant parameter μ , the log mean goal arrival times in away games is expected to be 4.025 which corresponds to around 56 min when two teams of average performance competing against each other.

Regarding the shape parameter γ , the data suggest a posterior mean of 1.13 and a 95% posterior interval ranging from 1.078 to 1.186. The posterior distribution of γ is clearly greater than one, implying (a) evidence against the exponential model (where $\gamma = 1$) and in favour of the Weibull model, and (b) that the goal-scoring rate increases as the match progresses. This result aligns with the findings reported by Silva and Swartz (2016), which indicate that goal scoring is not constant during a match. Moreover, the Weibull model was also found better than the exponential one in terms of DIC (11,010 vs. 11,040, respectively); see also Sect. 3.2.1 and Table 2 for details.

Before proceeding with the analysis of the team ability parameters, we should note that the attacking abilities of teams in goal arrival time models are a decreasing function of their actual attacking strength. Better teams tend to score more frequently, meaning they have shorter time intervals between their goals. Consequently, the Weibull ability parameters will be lower for stronger teams. Equivalently, the defensive team parameters will be an increasing function of each team's defensive quality, indicating that teams with better defensive skills will delay the scoring time of their opponents.

Figure 1 presents the 95% posterior credible intervals of the team attacking and defensive ability parameters (att_k and def_k for $k = 1, \dots, 20$) of the Weibull model. The model appears to capture in a reasonable way the attacking and defensive team abilities following their final league rankings. From this figure, we observe that Manchester City, Liverpool, Arsenal, Tottenham, and Manchester United not only have the lowest (i.e., the best) attacking abilities (sorted from smallest to largest), but their 95% posterior intervals do not include the value of zero, indicating their superiority over a team with "average" attacking ability. This pattern does not exactly match the final league table rankings, as these teams finished 1st, 2nd, 5th, 4th, and 6th. However, it aligns perfectly with their ranking by the number of goals scored. For this reason, although Chelsea finished 3rd in the league, it has the 6th-best attacking ability, as this was Chelsea's ranking in terms of goals scored. In terms of defence, Manchester City and Liverpool are once again dominant, followed by Chelsea, with the highest defensive ability parameters. For these three teams, the posterior distributions of the defensive parameters are well placed above zero, indicating a clearly greater defensive performance over a team with "average" defensive ability. Regarding the below 'average' worst-performing teams, Huddersfield, Fulham, Cardiff, and Brighton had the poorest attacking abilities, while Fulham, Huddersfield, Bournemouth, Cardiff, Burnley, and Southampton had the weakest defensive abilities (from worst to best).

4.3 League reconstruction

Assessing the in-sample predictive ability of our model was what naturally followed after the previously mentioned clear indication of our model's decency. The predictive distribution was used to conduct such model assessment through MCMC sampling. In particular, we used the predictive distribution to regenerate the League's rank under our model, and make comparisons to the actual rank. This was a two-step procedure since we first had to simulate goal arrival times from our model's predictive distribution; see Appendix B, available in the electronic supplement. The pseudocode used for this purpose is presented in Algorithm 1 while the rationale behind this algorithm is presented in Appendix B, available in the electronic supplement.

Algorithm 1 Algorithm for league regeneration using weibull survival model

- **Input:** $(\gamma^{(t)}, \eta_{1,G}^{(t)}, \eta_{2,G}^{(t)})$: MCMC values of the parameters of the model for each MCMC iteration t and games G .
- **Output:** $L^{(t)} = Res[,]$
 - for $t = 1$ to T_{MCMC} do
 - $L^{(t)} = 0$
 - for $G = 1$ to N_G do
 - * $t_{inj} \sim \mathcal{U}[3, 7]$
 - * $gt = g_1 = g_2 = 0$;
 - * while($gt < 90 + t_{inj}$) do
 - for $k = 1$ to 2 do
 - $st_k \sim Weibull\left(e^{-\gamma^{(t)}\{\eta_{k,G}^{(t)} - \log \Gamma(1+1/\gamma^{(t)})\}}, \gamma^{(t)}\right)$
 - $gt = gt + \min(st_1, st_2)$
 - if ($gt > 90 + t_{inj}$) stop
 - if ($st_1 < st_2$) $g_1 = g_1 + 1$; else $g_2 = g_2 + 1$
 - $Res[G,] = (g_1, g_2)$
 - $L^{(t)} = Res[,]$

Return League results $L^{(t)}$ for $t = 1, 2, \dots, T_{MCMC}$

Indexes: $G = 1, 2, \dots, N_G$; N_G : number of games $t = 1, 2, \dots, T_{MCMC}$; T_{MCMC} : number of MCMC iterations

Table 5 contains posterior, median, standard deviations and the 95% predictive credible intervals for the points of each team under the fixed effects Weibull model. In terms of the finally predicted rank, it is evident that the model exactly captures 10 out of 20 while all other actual points are within the 95% posterior credible interval of the predicted points. This result points to an outstanding agreement between our model and the actual rank since the structural assumptions of our model are minimal and the main modeling target is the scoring goal arrival times rather than the actual goals. More specifically, the model captures the top two highest-rated teams and the four lowest-rated teams accurately. All other deviations between our model and reality are minor since the maximum misplacement of a team in the final rank is observed is two ranks and the

Table 5 Reconstructed league for the EPL 2018–2019 data under the Weibull model

	Ranking		Obs	Posterior predicted points				
	Obs	Pred	Points	Median	Mean	SD	$P_{2.5\%}$	$P_{97.5\%}$
Man. City	1	1	98	90	90.12	7.05	76	103
Liverpool	2	2	97	89	88.89	7.22	74	102
Chelsea	3	4	72	67	66.63	8.67	49	83
Tottenham	4	3	71	69	69.33	8.57	52	86
Arsenal	5	5	70	65	65.16	8.68	48	82
Man. Und	6	6	66	59	58.78	8.79	42	76
Wolves	7	8	57	51	51.35	8.66	35	69
Everton	8	10	54	55	55.09	8.70	38	72
West Ham	9	7	52	48	48.15	8.91	31	66
Leiceister	10	12	52	55	54.52	8.65	38	72
Watford	11	9	50	48	48.00	8.53	32	65
Crystal Palace	12	11	49	51	50.77	8.76	34	68
Newcastle	13	13	45	46	46.10	8.25	30	63
Bournemouth	14	14	45	44	44.32	8.65	28	62
Burnley	15	16	40	40	39.87	8.21	25	57
Southampton	16	15	39	42	41.82	8.28	26	59
Brighton	17	17	36	36	36.15	8.05	21	53
Cardiff	18	18	34	32	31.91	7.56	18	47
Fulham	19	19	26	28	28.18	7.40	15	44
Huddersfield	20	20	16	21	21.68	6.37	10	35

Obs.: Observed; Pred.: Predicted; SD: Standard deviation; $P_{a\%}$: $a\%$ percentile; Predicted ranking is based on posterior mean predicted points

difference between points can be translated as misspredicting a maximum of two matches. Overall, our model appears to be predicting the exact position of the top and worst teams in the league while making slight misspredictions in the medium-ability teams where the posterior ranking probabilities are too close to making clear ranking discrimination and thus adds to the plausibility of this parsimonious fixed effects model for in-sample predictions especially when the end-point is predicting the top/worst teams’ rankings.

Table 6 presents the Root Mean Square Error (RMSE) of the posterior estimates for the mean and median points from the reproduced final league table under the fixed-effects Weibull model. The results further validate the model’s goodness-of-fit, as it reproduces the final league standings with an average error of approximately four points and one ranking position. This indicates that, on average, the model maintains accurate predictions with an error, on average, of one win and one draw for each team.

Another useful insight drawn from Table 6 is that the RMSE for the posterior mean estimate of ranks is calculated to be 1.4 positions in the final league table. This implies that, on average, our model predicts final league positions with a

Table 6 Root mean square error (RMSE) of the posterior mean and median points along with the posterior mean of ranks after 12,000 replications of the League under the proposed model

	Mean points	Median points	Rank
RMSE	4.168	4.153	1.4

discrepancy of just one or two positions. This result further supports the model's strong fit to the data, particularly given the simplicity of the proposed fixed-effects model and its indirect approach to modeling the goals scored by each team.

4.4 Out-of-sample prediction

The out-of-sample predictive ability of the model is also of major scientific concern, especially due to its intriguing nature and its' applications to betting. In this case, the predictive ability of the model is expected to lessen compared to the in-sample measures presented in the previous subsections, however, the loss of out-of-sample predictive ability of a model with that impressive in-sample predictive performance is expected to be minor.

Under the Bayesian framework, the only adjustment required to assess the out-of-sample ability of our model is to assume that a portion of the data is known, and the rest of the data is simulated using the predictive distribution. This is to say that in the in-sample model assessment conducted previously, we assumed that the entirety of the data (League) is known and each new League was simulated from the posterior distribution under the assumption that the observed teams' abilities hold true for the teams' considered in the simulation of each new league.

4.4.1 Second half of the season prediction

The choice of having as training dataset the first half of the season can be reasonably justified in both purely statistical terms, and by the nature of the sport under the League considered. From a purely statistical point of view, having half the data as a training dataset can be considered to be sufficient in terms of predicting the other half of the data since we are using half of the entirety of the information contained in the original dataset for learning purposes. The additional argument from the nature of the sport and the League is that during the first half of the Season, each team is playing against each other team. That means that all teams have competed with each other at least once and hence if the endpoint was to maximize the amount of information contained in the training dataset while minimizing the length of the data used for its construction, that would be during the half of the English Premier League season.

Among the measures that assess the out-of-sample predictive ability of our model is the percentage of agreement in predicting the games of the second half of the season. Since Bayesian theory allows us to make inference on the entirety

of its distribution of that percentage, and not consider it as a point estimate, the posterior distribution presented on the left panel of Fig. 2, contains all the information we need to get an initial idea of how are model performs. More specifically, the posterior median (mean) is 46.84% (46.77%) with a 95% credible interval of 40.53% – 53.16%. Those posterior measures are indications that our model performs better than expected especially considering the fact that if we were to choose the outcome of matches at random, we would expect to make accurate predictions in 33% of the time. Since that percentage is not included in the 95% credible interval, our parsimonious fixed effects model with minimum assumptions, provides significantly better outcome prediction even in the out-of-sample case.

This results can be made more comprehensible if we consider two events instead of three, thus making comparisons with a percentage of 50% which is the expected times of correct predictions if we were to choose between two outcomes completely at random. Therefore, if we assume that Win or Loss of a match is one event and draw is the other event, then the posterior distribution presented on the right panel of Fig. 2 reveals very promising results with regards to the predictive ability of our model making it more easily understandable since the posterior median (mean) are now 70.53% (70.45 %) with a 95% credible interval of 64.74–75.79%.

4.4.2 Brier score

Another useful metric for measuring the predictive ability of our model by understanding match-specific surprising outcomes is the Brier penalty (BP). This metric has been used by Spiegelhalter and Ng (2009) as a modification to the original Brier Score introduced by Brier (1950), and can be expressed as the squared distances between a vector of probabilities and the actual outcome.

The Brier Score for categorical data is given by

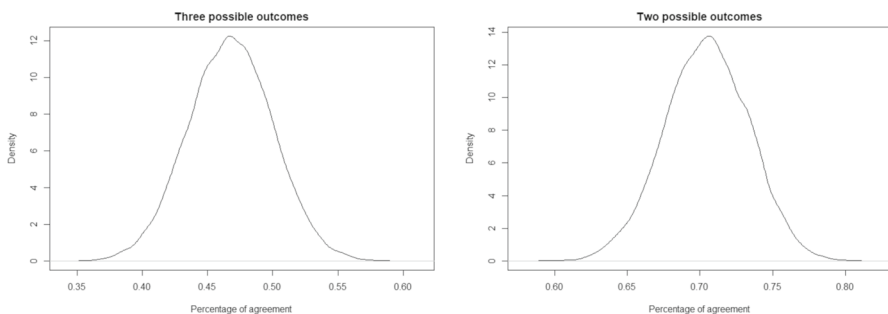


Fig. 2 Posterior distributions of the percentage of the agreement in predicted games for the second half of the season for English Premier League 2018/19 under the proposed fixed effects Weibull model. The posterior distribution of the percentage of agreement on the left applies when the events of interest are three (aka Win/Draw/Loss) while on the right, the posterior distribution applies when the event of interest is considered to be two (Win & Loss/Draw)

$$BS = \frac{1}{N} \sum_{k=1}^N \left[\sum_{j=1}^J (p_k(j) - a_{kj})^2 \right],$$

where N is the number of predicted outcomes, J is the number of categories, $p_k(j)$ is the predicted probability for k outcome and j category, and a_{kj} is the binary variable taking the value of one if category j observed and zero otherwise.

In our case, we consider the categorical variable Y_i denoting the final outcome of game i (for $i = 1, \dots, N$) in the form of (home) win ($Y_k = 1$), draw ($Y_k = 2$) and loss ($Y_k = 3$); where N denotes the number of games under consideration. Hence, $\mathbf{a}_k = (a_{k1}, a_{k2}, a_{k3})$ will be equal to $(1, 0, 0)^T$ for a home win, $(0, 1, 0)^T$ for a draw, and $(0, 0, 1)^T$ for the home loss. We can rewrite the Brier score as

$$BS = \frac{1}{N} \sum_{k=1}^N BS_k \text{ and } BS_k = \sum_{j=1}^3 (p_k(j) - a_{kj})^2.$$

The Brier Score for k game decomposes to

$$\begin{aligned} BS_k &= \sum_{j=1}^3 \left(p_k(j)^2 + a_{kj}^2 - 2a_{kj}p_k(j) \right) \\ &= \sum_{j=1}^3 p_k(j)^2 + 1 - 2p_k(Y_k) \end{aligned}$$

This expression is exactly the modified Brier Score named as Brier Penalty (BP) in Spiegelhalter and Ng (2009) which is association football (soccer) can be written in the following simplified form:

$$BP_k = 1 + P(\text{Home Win})^2 + P(\text{Draw})^2 + P(\text{Away Win})^2 - 2P(\text{Actual Outcome}).$$

All probability outcomes in the above formula refer to the k -th game. The support of this metric is $[0, 2]$ with $BP \rightarrow 0$ indicating perfect predictions while $BP \rightarrow 2$ indicating the opposite.

We applied this metric to all 190 matches simulated from the predictive distribution obtaining a Brier Penalty for each of them. Even though the bibliography regarding this metric is unclear with regards to the threshold above which the model's predictive ability can be problematic, we concentrated our attention on matches for which the Brier Penalty was above 1 to account for the symmetry of this metric interpretational property.

One direct positive insight after computing this metric was that only 31 out of a total of 190 matches have gotten a Brier Penalty of above 1. This meant that our model provided decent out-of-sample predictions for around 83.7% of cases hence providing more evidence that adds we introduced an adequate predictive model in both the in-sample and the out-of-sample case.

Since the Brier Penalties can be calculated for each match, it can be insightful in investigating surprising results in specific matches of the League. In our case, these matches are presented in Table 7 along with the final rank difference of the

teams participating. The ranking differences presented in Table 6 act as a tool for making sense of the matches that our model did not capture. In particular, Table 6 clearly indicates that those miss-predictions can be justified by the fact that in most cases, the final outcome is the opposite of what would have logically been predicted if we were to use the final ranks of teams for prediction of outcomes. The most obvious example of the existence of this phenomenon is the match between Newcastle

Table 7 Assessed probabilities for 31 score-specific matches of the second half of the season with Brier penalties above one, using the Weibull model

Match opponents		Final score	Probability			Rank difference
			Home		Away	
Home	Away		Win	Draw	Win	
Tottenham	Wolves	1–3	0.69	0.19	0.11	−3
Leicester	Cardiff	0–1	0.65	0.19	0.15	−9
Burnley	West Ham	2–0	0.20	0.21	0.59	5
Chelsea	Southampton	0–0	0.79	0.13	0.08	−13
Tottenham	Man. Und	0–1	0.68	0.16	0.16	−2
Leicester	Southampton	1–2	0.60	0.22	0.18	−7
Watford	Bournemouth	0–0	0.67	0.18	0.15	−4
Bournemouth	Chelsea	4–0	0.17	0.18	0.65	11
Man. Und	Burnley	2–2	0.80	0.11	0.09	−9
Newcastle	Man City	2–1	0.05	0.13	0.82	12
Liverpool	Leicester	1–1	0.77	0.17	0.06	−7
West Ham	Liverpool	1–1	0.06	0.15	0.79	8
Brighton	Burnley	1–3	0.61	0.21	0.18	2
Man. Und	Liverpool	0–0	0.08	0.14	0.78	4
Burnley	Tottenham	2–1	0.05	0.10	0.85	11
Tottenham	Arsenal	1–1	0.58	0.19	0.23	−1
Everton	Liverpool	0–0	0.07	0.15	0.78	6
Southampton	Tottenham	2–1	0.10	0.13	0.77	12
Chelsea	Wolves	1–1	0.65	0.22	0.13	−4
Bournemouth	Burnley	1–3	0.66	0.17	0.17	−1
Leicester	Newcastle	0–1	0.56	0.28	0.17	−4
Fulham	Everton	2–0	0.16	0.17	0.67	11
Chelsea	Burnley	2–2	0.87	0.09	0.04	−12
Bournemouth	Fulham	0–1	0.65	0.17	0.17	−5
Arsenal	Crystal Palace	2–3	0.68	0.18	0.14	−7
Tottenham	West Ham	0–1	0.72	0.16	0.12	−6
Huddersfield	Man. Und	1–1	0.14	0.16	0.70	14
Bournemouth	Tottenham	1–0	0.14	0.16	0.70	10
Arsenal	Brigton	1–1	0.67	0.17	0.15	−12
Tottenham	Everton	2–2	0.69	0.16	0.15	−4
Man. Und	Cardiff	0–2	0.75	0.14	0.11	−12

United and Manchester City. The winner of this League lost to a team with a rank distance of 12 positions in the final rank with a score of 2–1. This surprising aspect can be observed for the majority of matches in Table 6 with 20/31 matches having an absolute ranking difference of above or equal to 5 ranks (with the magnitude of that metric being above 10 for 11 matches).

Moving beyond the specifics of the miss-predictions and their justifications, we compared the Brier penalties concerning the out-of-sample predictions under our model with the Brier penalties if we were to assign equal probabilities to each of the outcomes. That would mean that instead of assigning the Win/Draw/Loss probabilities according to our model, we would assign a probability of 0.33 to each, making it equivalent to comparing our model to random chance mimicking what the analysis we conducted in the in-sample case. According to Spiegelhalter and Ng (2009) this comparison can be made by summing over all Brier penalties calculated under our model and then comparing it to the summation of all Brier penalties calculated under random chance. The Brier penalty concerning the 190 matches of the second half of the Season under our model is calculated to be ≈ 105.35 whereas the Brier penalty under random chance is calculated to be ≈ 126.67 making which can be translated into a 16.82% difference in favour of our model.

Taking into consideration the information presented in this subsection, we can draw the safe conclusion that the miss-predictions of our proposed model, are justified since any discrepancies between what the model predicted in terms of probability that translate to a Brier penalty, and reality, are what it was a-prior expected provided information concerning the final ranking of the teams. Therefore, our model appears to be performing exceptionally well even in out-of-sample scenarios.

5 Further modeling issues

5.1 Comparison to the double poisson model

To further affirm both the validity of our methods, especially those concerning the transcription of the goal arrival times to actual goal events, and its in-sample predictive ability, our results concerning the regeneration of the EPL league were compared to a well-established and widely used model for modeling goals events as integers, the double Poisson model.

Comparing Tables 8 to 5, it is evident that both models seem to capture the final ranking of EPL in a compatible manner and very precise manner. To be more specific, both models seem to affirm that the median points of each team are systematically lower than the actual points while all actual points are captured accurately by each 95% credible interval. To provide further evidence of the surprising comparability of the two models, the percentage of ranks that are predicted exactly is 50%. This is a surprising finding given the fact that our model makes limited assumptions and models the actual events (goals) indirectly. A final surprising remark concerning the proposed model is that the posterior standard deviation of each team's points is systematically lower than that in the double

Table 8 Reconstructed league for the EPL 2018–2019 data under the double Poisson model

	Ranking		Obs points	Posterior predicted points				
	Obs	Pred		Median	Mean	SD	$P_{2.5\%}$	$P_{97.5\%}$
Man City	1	1	98	92	91.74	7.33	76	105
Liverpool	2	2	97	91	90.49	7.53	75	104
Tottenham	4	3	71	70	69.70	9.35	51	87
Chelsea	3	4	72	68	67.55	9.49	49	86
Arsenal	5	5	70	66	65.40	9.46	47	84
Man United	6	6	66	60	59.55	9.52	41	78
Everton	9	7	54	58	57.89	9.62	39	76
Leicester	7	8	52	55	54.70	9.66	36	74
Wolves	8	9	57	53	53.16	9.47	34	72
Crystal Palace	11	10	49	52	51.75	9.68	33	71
West Ham	12	11	52	51	51.25	9.61	33	70
Watford	10	12	50	49	49.22	9.60	31	68
Newcastle	13	13	45	48	48.52	9.55	30	68
Bournemouth	14	14	45	46	46.11	9.50	28	65
Southampton	16	15	39	41	41.51	9.21	24	60
Burnley	15	16	40	40	40.18	9.12	23	58
Brighton	17	17	36	37	37.21	8.96	21	55
Cardiff	18	18	34	32	32.32	8.50	17	50
Fulham	19	19	26	27	27.74	8.01	14	45
Huddersfield	20	20	16	21	21.41	7.13	9	37

Obs.: Observed; Pred.: Predicted; SD: Standard deviation; $P_{a\%}$: $a\%$ percentile; Predicted ranking is based on posterior mean predicted points

Poisson case, providing evidence that our model’s predictions are slightly more accurate with regard to the prediction of the rank of each team.

5.2 The red card effect

The impact of red cards on factors influencing goal-scoring rates has attracted scientific attention (Červený et al. 2018). Specifically, it is expected that an increase in the number of red cards received by a team reduces its probability of scoring. Consequently, the scoring time for that team should increase compared to the opposing team’s scoring time, assuming the latter does not also receive a red card during the same period. During the EPL 2018/19 season, a total of 47 red cards were issued across 380 matches. Subsequently, the red card covariate (x_{ij}) for period i and team j is incorporated into the model formulation in the following way:

$$x_{ij} = \begin{cases} 0, & \text{no red card during time period } i \\ -1, & \text{If the home team } (j = 1) \text{ plays with fewer players for period } i, \\ 1, & \text{If the away team } (j = 2) \text{ plays with fewer players for period } i \end{cases}$$

for $i = 1, 2, \dots, n$ and $j = 1, 2$. The values of x_{ij} reflect that when a team receives a red card, it is expected to negatively impact this team and positively benefit the opposing team. However, if no red card is issued during the match, it should have no impact on either team. If a red card is issued during period i , the covariate x_{ij} will be set equal to the proportion of time that the team received the red card plays with fewer players.

Here we consider two different models regarding the home effect. The first model assumes a common effect for both home and away teams while the the second assumes different home and away effects. Hence the red card indicator is inserted into the linear predictor of Model 1 as follows:

$$\begin{aligned} \log \{E(T_{i1})\} &= \mu + \text{home} + \text{att}_{HT_i} + \text{def}_{AT_i} + \beta x_{ij}, \\ \log \{E(T_{i2})\} &= \mu + \text{att}_{AT_i} + \text{def}_{HT_i} - \beta x_{ij} \end{aligned}$$

while in the second one β is substituted by β_1 and β_2 and hence the linear predictor is given by

$$\begin{aligned} \log \{E(T_{i1})\} &= \mu + \text{home} + \text{att}_{HT_i} + \text{def}_{AT_i} + \beta_1 x_{ij}, \\ \log \{E(T_{i2})\} &= \mu + \text{att}_{AT_i} + \text{def}_{HT_i} - \beta_2 x_{ij} \end{aligned}$$

Note the covariate (x_{ij}) is assigned a plus sign or a minus sign depending on whether the team that received the red card is the home team or the away team in order to capture the red card effect for both teams. For all red card coefficients, we have used a normal low information prior with a variance equal to 0.001. Posterior summaries for all red card effects are presented in Table 9 and Fig. 3.

For Model 1 (common red card effect), the negative posterior mean of this effect suggests that its presence decreases the goal arrival time for the team with the extra player. This is consistent with the intuition that the red card will benefit the team by playing with an extra player. However, this effect does not seem to be of importance since the 95% posterior interval includes the value of zero which corresponds to a no-card effect. Regarding Model 2, which assumes different red card effects for the home and away teams, the picture is slightly different since, although the home red card effect still seems to be of no importance, the posterior distribution for the

Table 9 Posterior summaries for the red card effects for the 2018–19 English premier league under the Weibull fixed effects model

Models	Red card effect	Mean	Median	SD	2.5%	97.5%
1	Common (β)	-0.259	-0.26	0.157	-0.571	0.039
2	Home (β_1)	-0.108	-0.117	0.204	-0.490	0.316
	Away (β_2)	-0.423	-0.421	0.209	-0.827	-0.021

Model 1: Common red card effect; Model 2: Separate red card effect for home and away teams

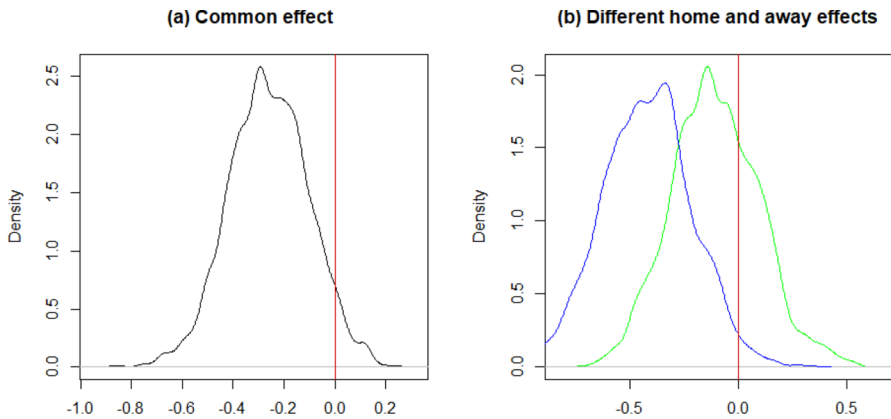


Fig. 3 Posterior densities for the red card coefficients for the models **a** with common effect **b** different home and away effects; the red vertical line at zero refers to no red card effect; MCMC details: 10,000 iterations, 1000 burn-in

away team suggests that the red card effect is away from zero having a major impact in their scoring times. Finally, the DIC for both models (common β and separate β 's) are the same and equal to the corresponding value of the Weibull fixed effects model.

5.3 Half-time censoring

A meaningful alternative to the data setup presented in Table 1 can be constructed assuming that the goal arrival times are not only bi-variately censored at the end of each match but also at half-time.

In practical terms, the alternative version of Table 1 can be constructed as:

Table 10 differentiates from Table 1 only in the sense that there is now an extra line in the data because of which time resets one additional time at the 45th minute of the regular time.

The validity of Tables 1 and 10 as data modeling structures, was tested by fitting the same fixed effects model for which the results were presented in Table 4 and comparing the DIC values, and it was found that the data fixed effects Weibull model under Table 10 received a DIC value of 11,070 which is higher than the DIC value for the same model but under the data structure of Table 1. This implies that the model fitted using Table 1 is preferable to the model using the data of Table 10.

5.4 Assessing the goal scoring rate

Although the main focus of this paper is on the expected goal arrival times, in this section, following a comment by a referee, we study and compare two additional models with an alternative structure on the scoring rate parameter γ . The first model considers different rate parameters γ for the home and away teams,

Table 10 Example of data layout assuming that both goal arrival times are censored at half (regular) time (45 min) for survival modeling: data refers to the first match of English premier league 2018–2019 (Manchester United – Leicester)

Game	Scoring time	Time interval	Home team	End-of-half censoring	Goal arrival times		Censoring times		
					Home	Away	Home	Away	
	T_i^*	T_i	$\mathcal{H}_i$	$\mathcal{C}_i$	t_{i1}	t_{i2}	c_{i1}	c_{i2}	
1	2	2	1	0	$\Rightarrow$	2	NA	0	2
1	NA	43	NA	1		NA	NA	43	43
1	82	37	1	0		37	NA	0	37
1	91	9	0	0		NA	9	9	0
1	NA	4	NA	0		NA	NA	4	4

while the second model assumes different shape parameters γ (or goal arrival rates) based on which team is ahead in the score. In this latter model, we use three parameters for γ : one for the leading team, one for when the game is tied, and one for the trailing team. Both of these models are compared with the standard model presented earlier in the paper, which assumes the same γ for both opponents.

Table 11 presents the estimated shape parameters γ which captures the goal scoring rate for the three different models under comparison. From this table, it is clear that the model assuming different goal-scoring rates for home and away teams is not notably different than the standard model with a common γ (DIC equal to 11,010 for both models). Additionally, the posterior distributions for the home and away γ are quite similar and they fall within a similar range as the common γ parameter of the standard model. The 95% posterior interval for the difference $\gamma_1 - \gamma_2$ ranges from -0.071 to 0.132 , with the value of zero located quite close to the centre of the posterior distribution. This result also indicates that this difference between home and away γ parameters is negligible and can be safely assumed to be equal to zero.

Table 11 Trailing team inducing variant scoring rate case

Model	Shape parameters	Posterior	Posterior	Posterior	Credible intervals		DIC
		Mean	Median	SD	2.5%	97.5%	
0	Common γ	1.131	1.130	0.028	1.078	1.186	11,010
1	Home γ_1	1.145	1.145	0.036	1.074	1.216	11,010
	Away γ_2	1.115	1.115	0.041	1.035	1.198	
2	Leading team (γ_1)	0.943	0.943	0.029	0.888	1.003	10,870
	No team ahead (γ_2)	1.044	1.042	0.050	0.951	1.145	
	Trailing team (γ_3)	1.731	1.727	0.076	1.588	1.889	

Model 0: Standard model with common γ ; Model 1: Model with different γ for home and away teams; Model 2: Model with different γ depending on which team is leading and which team is behind in score

On the other hand, Model 2 shows an improved DIC value compared to the standard model with a common γ (10,870 vs. 11,010), indicating that it better captures the overall rate of goal arrival times. Examining the posterior distributions (see Table 11), we observe that the posterior distributions of γ for the leading team, or when both teams are tied, are centered around one, suggesting a constant rate of goal arrival. However, for the trailing team, the posterior distribution of γ is clearly higher than one, confirming the findings of Silva and Swartz (2016), which reported that the trailing team is more likely to score the next goal.

6 Discussion

With the present paper, we are introducing the Weibull fixed effects model to the literature for modeling the gap scoring times of soccer data under a Bayesian survival analysis framework. The existence of the censored observations in the data is handled by the proposed structure of the data in order to be able to model them using the predictive distribution. The predictive distribution was also used for converting the goal arrival goal arrival times into actual goals under a first proposed simple algorithm based on discriminating between goal arrival times. The model proposed appears to provide a very satisfying fit to the data since it appears to be able to replicate the final ranking of the teams in a precise manner. Additionally, the model appears to be performing exceptionally well in both in-sample and out-of-sample predictions especially considering its' simplicity and the fact that it requires minimal assumptions with regards to the scoring goal arrival times that it models.

Even though the finally proposed model appears to have an overwhelming fit and predictive ability, further extensions like simultaneous modeling of the goal arrival goal arrival times with the use of a bi-variate distribution that captures both positive and negative correlations between the scoring times may lead to further predictive ability increase. However, assessing both the bivariate Marshall Olkin model and the random effects models, did not yield to a satisfactory model fit.

In our proposed modeling approach, the total number of goals (and intervals/events) n is unknown prior to observing the game. Consequently, our analysis is conditional on the observed number of goals. This approach is used for practical reasons in order to apply a simplified modeling approach. The implications of this assumption should be further investigated in future work by considering more sophisticated modeling approaches. One potential approach could involve extending the model by introducing an additional hierarchical layer; however, this would necessitate a considerably more sophisticated method for modeling the survival times. Given the complexity of this task, we have not pursued this issue further in this work.

Although in this paper we have worked with the use of low-information prior distributions, resulting in an analysis of an "objective" fashion where Bayesian inference relies solely on data, in football, sports fans and experts often possess good knowledge of the game. This expertise can be incorporated into informative prior distributions, potentially improving the model's precision. Nevertheless, eliciting such prior beliefs poses a challenge, as it requires transforming common knowledge into meaningful mathematical prior distributions. For instance, while experts might

easily rank the relative strength of teams, it would be considerably more difficult for them to express this knowledge in terms of scoring times in football matches. Alternatively, informative prior distributions might be constructed through the use of historical data and possibly the formulation of power priors (e.g. Ibrahim and Chen 2000). For example, Egidi et al. (2018) explored incorporating historical data into modeling by representing the scoring rates of each team as convex combinations of parameters estimated from betting odds and historical data.

Finally, further work is required for the incorporation of in-play covariates that add to the predictive power of any chosen model which can be achieved by applying Bayesian variable selection techniques (e.g. Dellaportas et al. 2002).

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References

- Andersen PK, Gill RD (1982) Cox's regression model for counting processes: a large sample study. *Ann Stat* 10(4):1100–1120
- Brier GW (1950) Verification of forecasts expressed in terms of probability. *Mon Weather Rev* 78(1):1–3
- Červený J, van Ours JC, van Tuijl MA (2018) Effects of a red card on goal-scoring in world cup football matches. *Empir Econ* 55:883–903
- Cox DR (1972) Regression models and life-tables. *J R Stat Soc Ser B (Methodological)* 34(2):187–220
- Dellaportas P, Forster JJ, Ntzoufras I (2002) On Bayesian model and variable selection using MCMC. *Stat Comput* 12(1):27–36
- Dixon M, Robinson M (1998) A birth process model for association football matches. *J R Stat Soc Ser D (The Statistician)* 47(3):523–538
- Dixon MJ, Coles SG (1997) Modelling association football scores and inefficiencies in the football betting market. *J R Stat Soc Ser C (Applied Statistics)* 46(2):265–280
- Egidi L, Gabry J (2018) Bayesian hierarchical models for predicting individual performance in soccer. *J Quant Anal Sports* 14(3):143–157
- Egidi L, Pauli F, Torelli N (2018) Combining historical data and bookmakers' odds in modelling football scores. *Stat Model* 18(5–6):436–459
- Hvattum LM (2017) Ordinal versus nominal regression models and the problem of correctly predicting draws in soccer. *J Homepage* 16(1):50–64
- Ibrahim JG, Chen MH (2000) Power prior distributions for regression models. *Stat Sci* 15:46–60
- Karlis D, Ntzoufras I (2000) On modelling soccer data. *Student* 3(4):229–244
- Karlis D, Ntzoufras I (2003) Analysis of sports data by using bivariate Poisson models. *J R Stat Soc Ser D (The Statistician)* 52(3):381–393
- Karlis D, Ntzoufras I (2009) Bayesian modelling of football outcomes: using the Skellam's distribution for the goal difference. *IMA J Manag Math* 20(2):133–145
- Kleinbaum DG, Klein M (2012) *Survival analysis: a self-learning text*, 3rd edn. Springer, Cham
- Koopman SJ, Lit R (2015) A dynamic bivariate Poisson model for analysing and forecasting match results in the English premier league. *J R Stat Soc Ser A (Statistics in Society)* 178:167–186
- Maher MJ (1982) Modelling association football scores. *Stat Neerl* 36(3):109–118
- Nevo D, Ritov Y (2013) Around the goal: examining the effect of the first goal on the second goal in soccer using survival analysis methods. *J Quant Anal Sports* 9(2):165–177
- Owen A (2011) Dynamic Bayesian forecasting models of football match outcomes with estimation of the evolution variance parameter. *IMA J Manag Math* 22(2):99–113

- Rue H, Salvesen O (2000) Prediction and retrospective analysis of soccer matches in a league. *J R Stat Soc Ser D (The Statistician)* 49(3):399–418
- Silva RM, Swartz TB (2016) Analysis of substitution times in soccer. *J Quant Anal Sports* 12(3):113–122
- Spiegelhalter D, Ng YL (2009) One match to go! *Significance* 6(4):151–153
- Spiegelhalter DJ, Best NG, Carlin BP et al (2002) Bayesian measures of model complexity and fit. *J R Stat Soc Ser B (Statistical Methodology)* 64(4):583–639
- Štrumbelj E, Šikonja MR (2010) Online bookmakers' odds as forecasts: the case of European soccer leagues. *Int J Forecast* 26(3):482–488
- Thomas AC (2007) Inter-arrival times of goals in ice hockey. *J Quant Anal Sports*. <https://doi.org/10.2202/1559-0410.1064>
- Tsokos A, Narayanan S, Kosmidis I et al (2019) Modeling outcomes of soccer matches. *Mach Learn* 108(1):77–95

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